

Juniper Networks® vMX Release Notes

Release 18.1R2
21 February 2019

These release notes accompany this release of the Juniper Networks® virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

Contents

Introduction	2
New and Changed Features	2
Known Behavior	2
Known Issues	2
Subscriber Management	3
Upgrade Instructions	3
System Requirements	3
Minimum Hardware and Software Requirements for KVM	4
Minimum Hardware and Software Requirements for VMware	6
Verifying Security Signatures	8
Requesting Technical Support	10
Self-Help Online Tools and Resources	10
Creating a Service Request with JTAC	10
Revision History	11

Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We recommend Ubuntu as the host OS.

vMX allows you to leverage Junos OS Release 18.1 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

New and Changed Features

This section describes the new features and enhancements in this release.

- **DPDK 17.08 support**—vMX uses DPDK Release 17.08 to support improved performance and enhancements such as DPDK poll mode drivers.

For optimal performance, we recommend 1G Huge Pages and 16G memory for each NUMA socket. You must set up Huge Pages when preparing the Ubuntu host.

- **Dual Routing Engines support for KVM**—You can configure dual Routing Engines on the same server or across two servers on KVM. You can deploy vMX with two VCP instances on separate hosts if the servers have Layer 2 connectivity. The VFP instance is mapped to the VCP instances on the two hosts.

To support this configuration, you must configure an additional VCP instance in the `vmx.conf` file and designate which instance is deployed on the server.

Known Behavior

This section contains the known behaviors and limitations in this release.

- Scale limitation is observed with VLAN tag operation and circuit cross-connect (CCC).
- When vMX is deployed, the management port is not configured, so you must use the serial console for configuration. Only a small number of configuration lines can be pasted in the vMX console. As a workaround, perform initial configuration to set the root password and to allow SSH access in the vMX console and perform further configuration using SSH.
- ICMP echo request packets are handled inline, which means ICMP packets do not reach the VCP and are replied to by the VFP itself. No ICMP packets are seen on the VCP and packet capture tools do not capture ICMP packets on the VCP.

Known Issues

This section lists the known issues in this release.

- When the FPC is restarted, **kernel: GENCFG: op 32 (Resync blob) failed** syslog message appears. This harmless message can be ignored. PR1050467
- LACP packets are getting dropped on the bridge. PR1059231
- LLDP packets are getting dropped on the bridge (for virtio). PR1066850
- When the FPC is restarting, **COS(cos_chassis_scheduler_pre_add_action:2137)** error messages appear indicating that flexible queuing mode is not enabled. PR1070655
- When the FPC is restarting, a **Received unsupported pic_mask 0x1 ignored message** message appears in the syslog file. PR1072436
- Traffic loss occurs at a remote receiver because of lost remote PIM joins to the local receiver. PR1087031
- When committing the configuration, **%DAEMON-4-SNMP_EVLIB_FAILURE: PFED ran out of transfer credits with PFE.Failed to get stats.** messages sometimes appear in the syslog file. PR1124902
- When committing the configuration, **pfed: %USER-3: downward spike received from pfe for opackets_reply:3545 opackets_record:48932521** messages sometimes appear in the syslog file. PR1146002
- Many **clksync_update_tod_to_pfe:109 Failed to send non-PEC pfe TOD update to other PFEs. Error code: 29.** messages appear in the syslog file. PR1148344
- Scheduler slips are occurring in large VPLS configurations and displaying **JTASK_SCHED_SLIP** log messages. PR1153290
- Observing **rpd[7116]: %DAEMON-6: ifl_delete: ifl** error messages that indicate a multicast tunnel (mt) has been deleted. PR1156725
- CPU pinning is not working properly with OVS bridges. PR1171096

Subscriber Management

- When bringing up an LNS session on vMX, the Cos-Shaping-Rate attribute (ERX-Attr-177) is sometimes omitted from the Acct-Start messages sent to RADIUS. PR1167154

Upgrade Instructions

You cannot upgrade Junos OS for the vMX router from earlier releases using the **request system software add** command.

You must deploy a new vMX instance using the downloaded software package.

System Requirements

These topics provide system requirements for each supported environment.

- [Minimum Hardware and Software Requirements for KVM on page 4](#)
- [Minimum Hardware and Software Requirements for VMware on page 6](#)

Minimum Hardware and Software Requirements for KVM

Table 1 on page 4 lists the hardware requirements.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Sample system configuration	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type, use Intel 82599-based PCI-Express cards (10 Gbps) and Ivy Bridge processors.
Number of cores	For lite mode with lab simulation use case applications: Minimum of 3 NOTE: Performance mode is the default mode and the minimum value is based on one port. • 1 for VCP • 2 for VFP NOTE: If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode. For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. NOTE: To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—$(4 * \text{number-of-ports}) + 4$ • With CoS—$(5 * \text{number-of-ports}) + 4$ NOTE: All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.

Table 1: Minimum Hardware Requirements for KVM (continued)

Description	Value
Memory	For lite mode: Minimum of 3 GB
NOTE: Performance mode is the default mode.	• 1 GB for VCP • 2 GB for VFP For performance mode: • Minimum of 5 GB 1 GB for VCP 4 GB for VFP • Recommended of 16 GB 4 GB for VCP 12 GB for VFP Additional 2 GB recommended for host OS
Storage	Local or NAS
	Each vMX instance requires 30 GB of disk storage
Other requirements	Intel VT-d capability
	Hyperthreading (recommended)
	AES-NI

[Table 2 on page 5](#) lists the software requirements for Ubuntu.

Table 2: Software Requirements for Ubuntu

Description	Value
Operating system	Ubuntu 14.04.1 LTS (recommended host OS) Linux 3.13.0-32-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11
Required packages	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev
NOTE: Other additional packages might be required to satisfy all dependencies.	NOTE: libvirt 1.2.19

[Table 3 on page 5](#) lists the software requirements for Red Hat Enterprise Linux.

Table 3: Software Requirements for Red Hat Enterprise Linux

Description	Value
Operating system	Red Hat Enterprise Linux 7.2 Linux 3.10.0-327.4.5

Table 3: Software Requirements for Red Hat Enterprise Linux (continued)

Description	Value
Virtualization	QEMU-KVM 1.5.3
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools
NOTE: SR-IOV requires these packages: kernel-devel gcc	NOTE: libvirt 1.2.17 or later

Table 4 on page 6 lists the software requirements for CentOS.

Table 4: Software Requirements for CentOS

Description	Value
Operating system	CentOS 7.2 Linux 3.10.0-327.22.2
Virtualization	QEMU-KVM 1.5.3
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools
	NOTE: libvirt 1.2.19
	To avoid any conflicts, install libvirt 1.2.19 instead of updating from libvirt 1.2.17.

Minimum Hardware and Software Requirements for VMware

Table 5 on page 7 lists the hardware requirements.

Table 5: Minimum Hardware Requirements for VMware

Description	Value
Number of cores NOTE: Performance mode is the default mode and the minimum value is based on one port.	For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. NOTE: To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—$(4 * \text{number-of-ports}) + 4$ • With CoS—$(5 * \text{number-of-ports}) + 4$ NOTE: All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX. For lite mode: Minimum of 3 • 1 for VCP • 2 for VFP NOTE: If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.
Memory NOTE: Performance mode is the default mode.	For performance mode: • Minimum of 5 GB 1 GB for VCP 4 GB for VFP • Recommended of 16 GB 4 GB for VCP 12 GB for VFP For lite mode: Minimum of 3 GB • 1 GB for VCP • 2 GB for VFP
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage

Table 6 on page 8 lists the software requirements.

Table 6: Software Requirements for VMware

Description	Value
Hypervisor	ESXi 5.5 Update 2
Management Client	vSphere 5.5 or vCenter Server

Verifying Security Signatures

The vMX image is securely signed, so you can verify the signature for the image.

To verify the signature:

1. Download the public key certificate and the software image from the Juniper Networks [Download Software](#) page.
2. Import the public key into a temporary GPG key ring.

```
mkdir temp
cd temp
gpg --homedir . --import certificate-file
```

For example:

```
gpg --homedir . --import Juniper_vMX_public_key_2017.asc
gpg: WARNING: unsafe permissions on homedir `.`
Warning: using insecure memory!
gpg: keyring `./secring.gpg' created
gpg: keyring `./pubring.gpg' created
gpg: ./trustdb.gpg: trustdb created
gpg: key CA6E E4DD E89A EB4C F22F 6897 7B82 9893 BA75 0B9B: public key "vMX
Trusted Kernel 2017 <ca@juniper.net>" imported
gpg: Total number processed: 1
gpg:             imported: 1 (RSA: 1)
```



NOTE: The hexadecimal key value and name vary with each key. The hexadecimal value is an easy way to confirm that the key is authentic.

3. Verify the validity of the signature.

```
gpg --homedir . --verify signature-file file-to-be-signed
```

For example:

```
gpg --homedir . --verify vmx-17.2R1.4-1.tgz.sig jinstall-vmx-17.2R1.4-domestic-signed.tgz
```



```
gpg: WARNING: unsafe permissions on homedir `.'  
Warning: using insecure memory!  
gpg: Signature made Tue Jun  6 16:50:05 2017 PDT using RSA key ID BA750B9B  
gpg: Good signature from "VMX Trusted Kernel 2017 <ca@juniper.net>"  
gpg: This key is certified with a trusted signature!  
gpg: WARNING: This key is not certified with a trusted signature!  
gpg:          There is no indication that the signature belongs to the owner.  
Primary key fingerprint: CA6E E4DD E89A EB4C F22F  6897 7B82 9893 BA75 0B9B
```

The second warning message appears because the key has not been marked as trusted. You can ignore the warning messages.

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or Partner Support Service support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at <https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf>.
- Product warranties—For product warranty information, visit <http://www.juniper.net/support/warranty/>.
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: <https://www.juniper.net/customers/support/>
- Search for known bugs: <https://prsearch.juniper.net/>
- Find product documentation: <https://www.juniper.net/documentation/>
- Find solutions and answer questions using our Knowledge Base: <https://kb.juniper.net/>
- Download the latest versions of software and review release notes: <https://www.juniper.net/customers/csc/software/>
- Search technical bulletins for relevant hardware and software notifications: <https://kb.juniper.net/InfoCenter/>
- Join and participate in the Juniper Networks Community Forum: <https://www.juniper.net/company/communities/>
- Create a service request online: <https://myjuniper.juniper.net>

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: <https://entitlementsearch.juniper.net/entitlementsearch/>

Creating a Service Request with JTAC

You can create a service request with JTAC on the Web or by telephone.

- Visit <https://myjuniper.juniper.net>.
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see <https://support.juniper.net/support/requesting-support/>.

Revision History

21 February 2019—Revision 2, Junos OS Release 18.1R2—vMX

4 June 2018—Revision 1, Junos OS Release 18.1R2—vMX

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